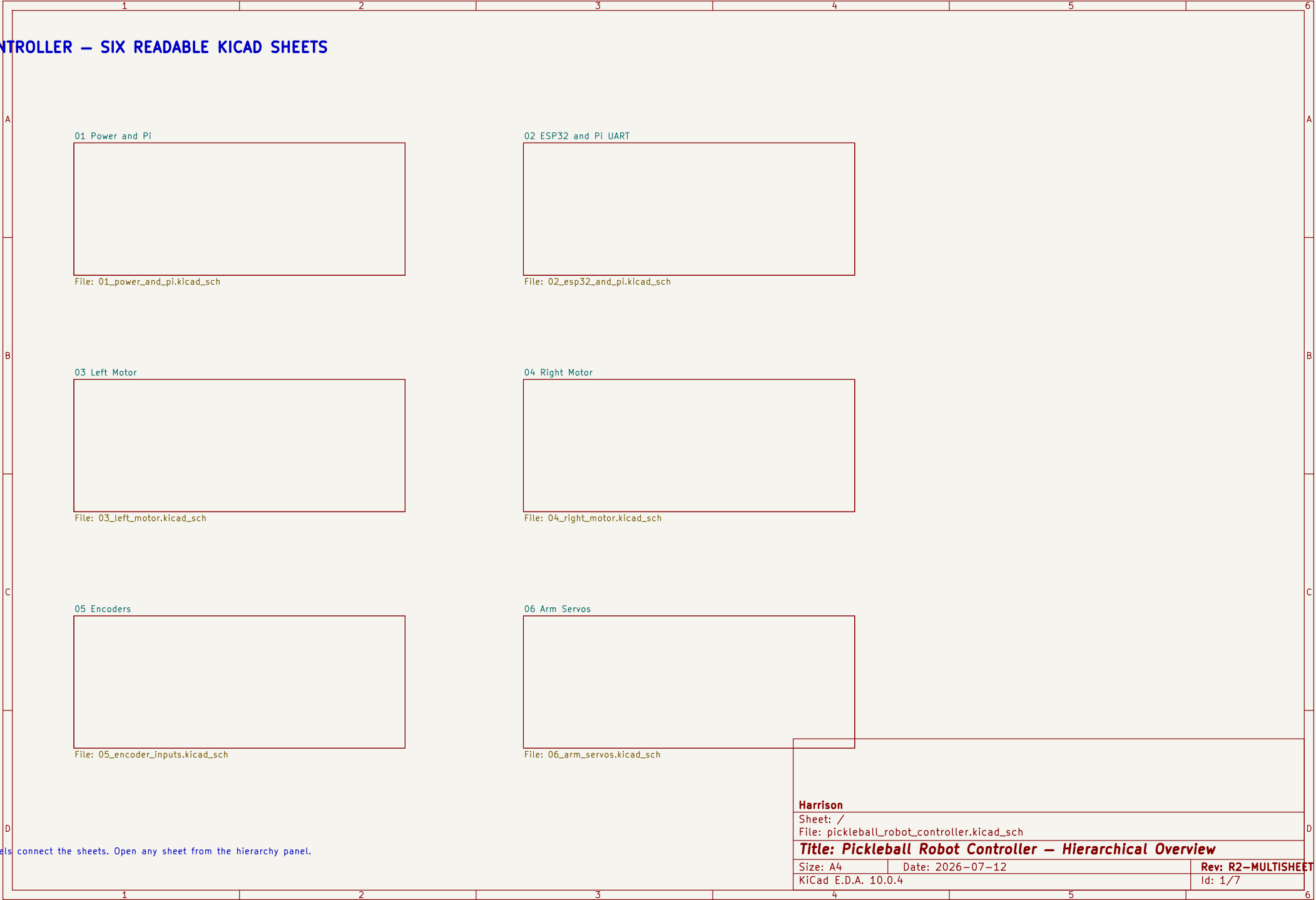


Robot Controller – Six Readable KiCad Sheets



net labels connect the sheets. Open any sheet from the hierarchy panel.

ERY INPUT AND PROTECTION

RASPBERRY PI 5 V / 5 A BUCK

external latching E-stop between F1 and the VBAT_SW distribution point.

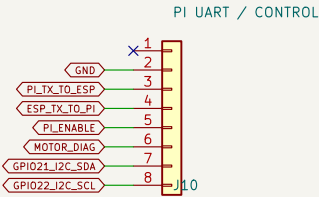
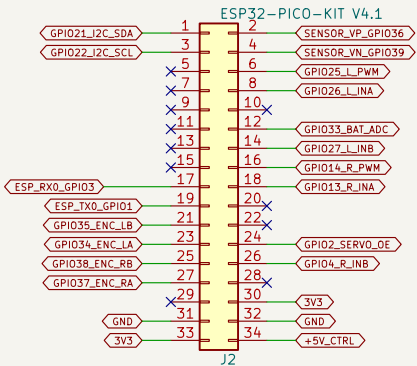
BATTERY VOLTAGE MONITOR

TROL BUCK – ESP32 AND LOGIC ONLY

PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG

CARRIER – CONFIRMED 17.78 mm ROW SPACING

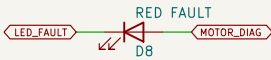
RASPBERRY PI UART / CONTROL – NORMAL 2.54 mm HEADER



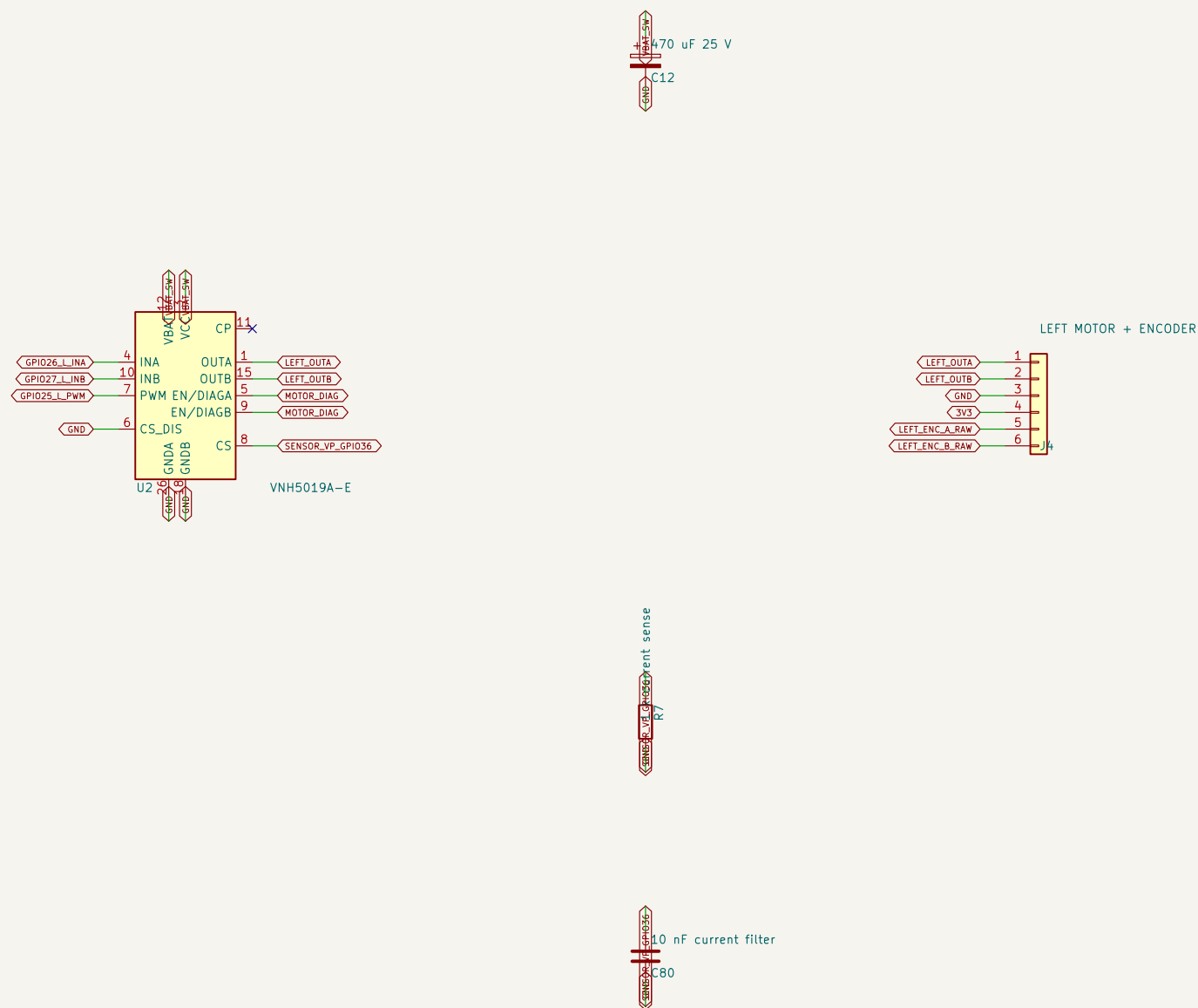
J10 pin 1 UNUSED; pin 2 GND; pin 3 Pi TX; pin 4 Pi RX; pins 5-8 optional control/I2C.

Pi GPIO14 physical pin 8 → J10.3; Pi GPIO15 physical pin 10 ← J10.4; Pi GND → J10.2.

STATUS LEDS

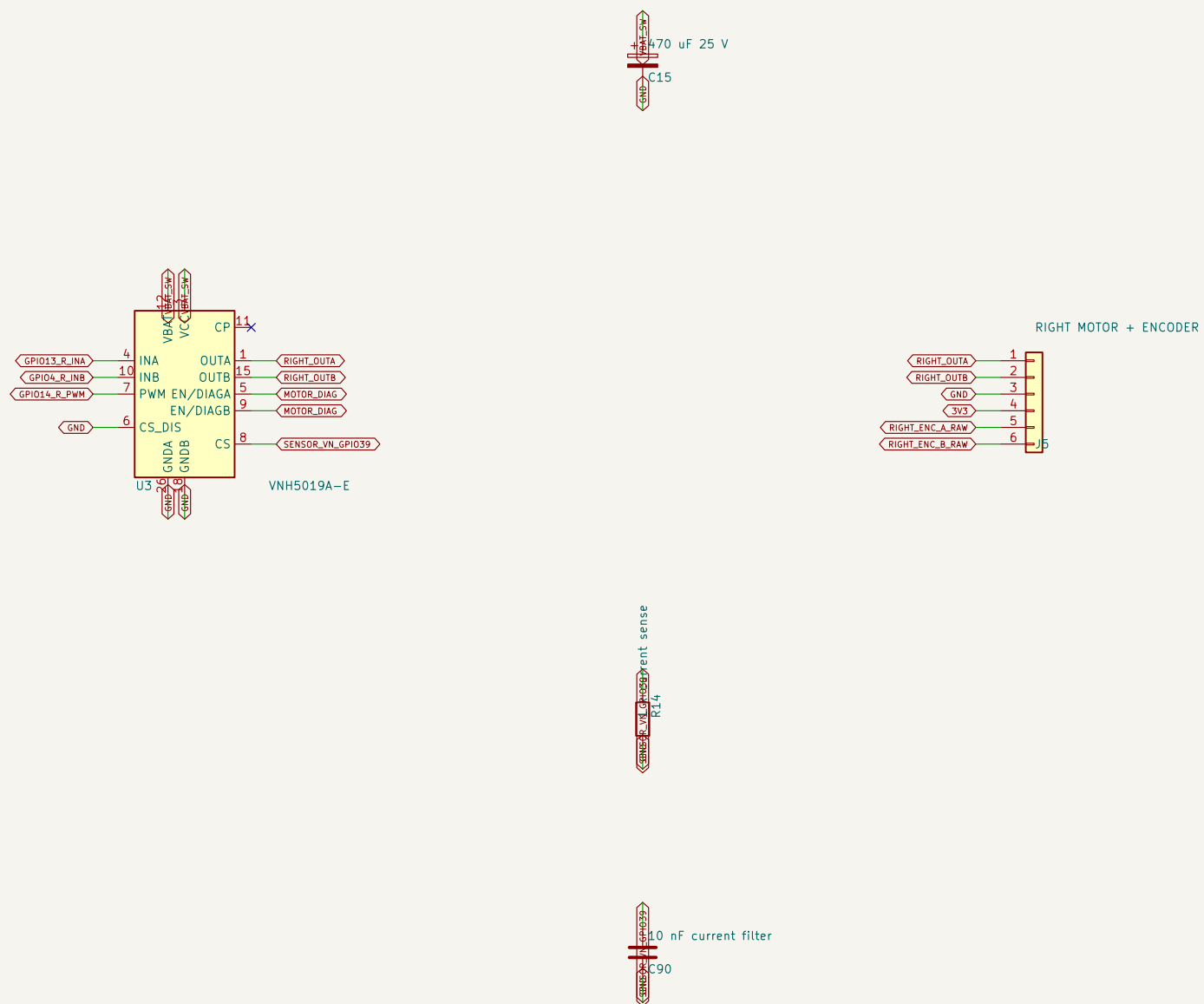


LEFT WHEEL MOTOR DRIVER



for A; 2 white motor B; 3 black GND; 4 blue +3V3; 5 yellow encoder A; 6 green encoder B.

RIGHT WHEEL MOTOR DRIVER



1 red motor A; 2 white motor B; 3 black GND; 4 blue +3V3; 5 yellow encoder A; 6 green encoder B.

SCHMITT-BUFFERED ENCODER INPUTS

A

B

C

D

tors are on the motor-driver sheets; their encoder A/B pins enter here.

F

A

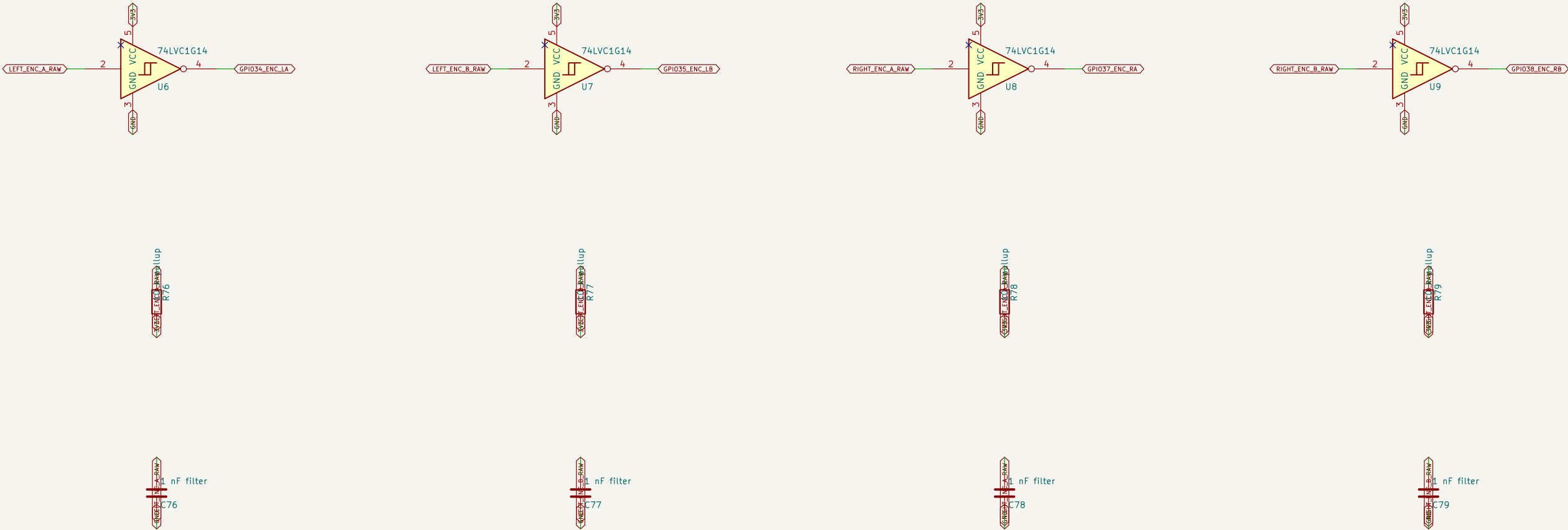
B

C

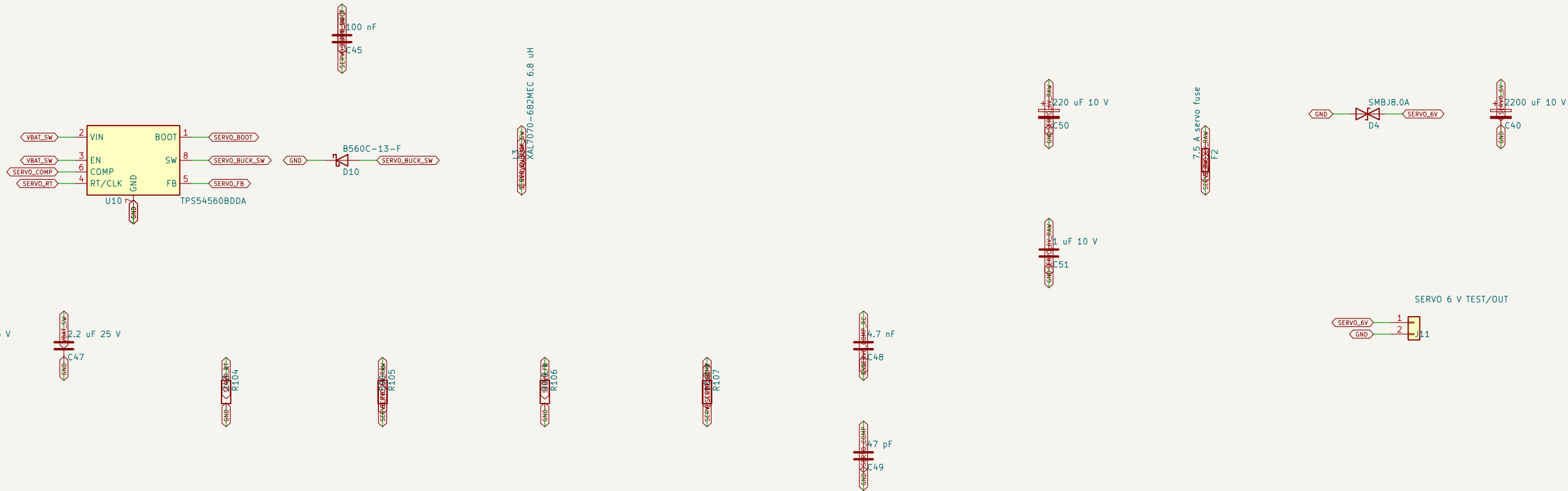
D

E

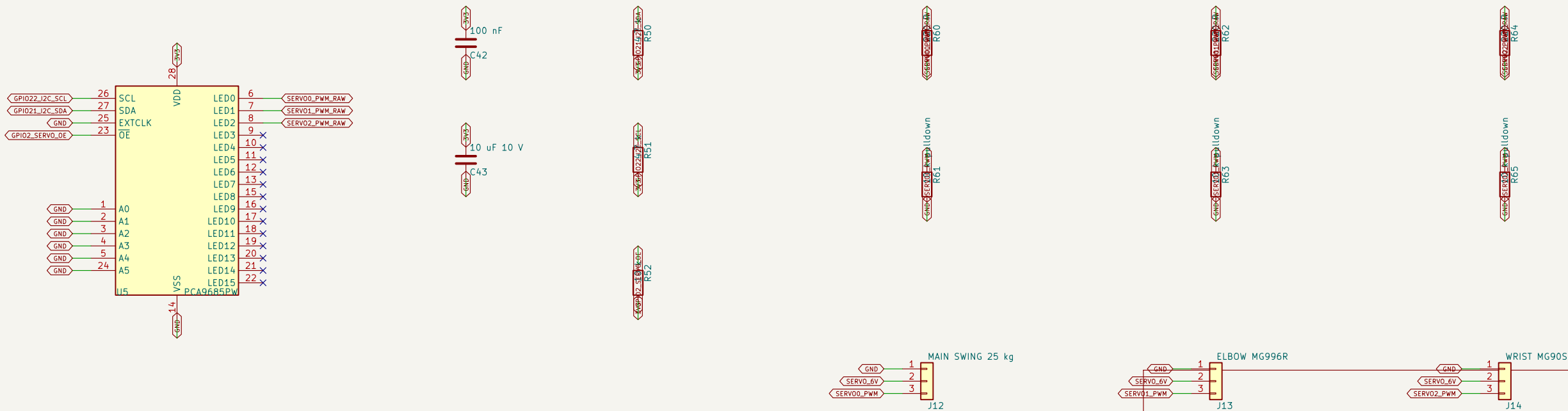
F



6 V / 5 A SERVO BUCK



A9685 SERVO PWM CONTROL



PWR_FLAG

Servo header order: pin 1 GND, pin 2 regulated +6 V, pin 3 PWM.

Sheet 6 of 6

Harrison

Sheet: /06 Arm Servos/

File: 06_arm_servos.kicad_sch

Title: Servo Power Regulator and 3-DOF Arm Control

Size: A3

Date: 2026-07-12

Rev: R2-MULTISHEET

KiCad E.D.A. 10.0.4

Id: 7/7